

Hypothesis Testing with E-Values

Chapter 13: FCR Control using E-Confidence Intervals

Based on Ramdas & Wang

Outline I

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Notation you'll need I

Recap: the classical confidence interval (CI). You already know this from intro stats. Data X is drawn from an unknown distribution $\mathbb{P}$, and we want to estimate a functional $\vartheta(\mathbb{P}) \in \Theta$ (e.g. a mean, a coin's bias, a median). A $(1 - \alpha)$ -CI is a data-dependent set $C(\alpha) \subseteq \Theta$ such that

$$\mathbb{P}(\vartheta(\mathbb{P}) \in C(\alpha)) \geq 1 - \alpha \quad \text{for every } \mathbb{P} \text{ under consideration.}$$

This is an *unconditional, pre-registered* guarantee: you must fix which parameter you're covering (and, implicitly, the level α) *before* looking at data. That is exactly the assumption we are about to break.

Notation you'll need II

E-confidence interval (e-CI), informally

An e-CI is a CI built by *inverting a family of e-values* instead of a family of p-values / test statistics. Concretely, suppose for every candidate value $\theta \in \Theta$ we have an e-variable $E(\theta)$ testing “ $\vartheta(\mathbb{P}) = \theta$ ” (i.e. $\mathbb{E}^{\mathbb{P}}[E(\theta)] \leq 1$ whenever $\vartheta(\mathbb{P}) = \theta$). Then

$$C(\alpha) := \left\{ \theta \in \Theta : E(\theta) < \frac{1}{\alpha} \right\}$$

is a $(1 - \alpha)$ -CI, by Markov's inequality – and we call it an e-CI. *Every* value of α can be read off from the *same* family of e-values, all at once.

Notation you'll need III

False coverage rate (FCR) – the selective-inference problem

Suppose you have K parameters $\theta_1^*, \dots, \theta_K^*$ and a $(1 - \alpha)$ -CI for each. If you look at the data, *select* a data-dependent subset $S \subseteq \{1, \dots, K\}$ of “interesting” ones (e.g. the largest effects), and only report CIs for S , the naive claim “these are 95% CIs” is generally **false**. The false coverage rate is

$$\text{FCR} := \mathbb{E}^{\mathbb{P}} \left[\frac{\sum_{i \in S} \mathbb{1}\{\theta_i^* \notin C_i(\alpha_i)\}}{|S| \vee 1} \right],$$

the *expected* fraction of reported intervals, among the selected ones, that miss their target.

Notation you'll need IV

The selection effect: why $\text{FCR} \neq \alpha$

Selecting “the most extreme” or “the most significant” parameters is not innocuous: extremeness in the *estimate* is correlated with the interval being pulled away from the truth (this is the same “winner’s curse” phenomenon behind regression to the mean). So the CIs you end up reporting are a biased sample of all the CIs you could have reported, and their conditional miscoverage rate can be much higher than the nominal α .

Notation you'll need V

The FDR analogy (Chapter 9). This is exactly the CI-analogue of the false discovery rate. Recall: if you test K hypotheses and only report the ones you *reject*, the fraction of those rejections that are false positives (FDP) can badly exceed α unless you correct for the fact that rejection is a data-dependent selection. FDR control (Benjamini–Hochberg, e-BH) fixes this for *point* null hypotheses. FCR control is the same story for *interval estimates*:

Chapter 9 (FDR)	Chapter 13 (FCR)
reject/don't-reject H_i	report/don't-report C_i
false positive: H_i true but rejected	false coverage: $\theta_i^* \notin C_i$ but reported
FDP = # false rejections / # rejections	FCP = # false coverages / # reported
e-BH procedure	e-BY procedure (this chapter)

The tool that makes the e-BY fix work cleanly is precisely the e-CI: because a family of e-CIs is valid *simultaneously at every level* α , you're allowed to choose α_i *after* seeing the data (in particular, depending on $|S|$) without breaking validity.

13.1 Intuition: why bother with e-CIs at all?

A classical $(1 - \alpha)$ -CI is built for one fixed α . If you want a 90% CI *and* a 99% CI, in general you need two separate constructions (or a union bound linking them).

An e-CI is built from a family of e-variables $\{E(\theta)\}_{\theta \in \Theta}$ testing every possible parameter value. Markov's inequality then gives you a valid CI *at every level α simultaneously*, for free, from that one object:

$$C(\alpha) = \{\theta : E(\theta) < 1/\alpha\}.$$

This “all levels for the price of one” property (called *level-free*) is exactly the extra structure that will let us handle a data-dependent selection rule in Section 13.2, and let us merge many uncertainty sets in Section 13.3.

Definition 13.1: families of e-variables and e-CIs I

Let $\mathcal{P}$ denote the set of all possible data distributions, $\vartheta : \mathcal{P} \rightarrow \Theta$ a functional of interest, and $\mathcal{P}_\theta := \{\mathbb{P} \in \mathcal{P} : \vartheta(\mathbb{P}) = \theta\}$ the distributions with that functional value.

Family of e-variables

$\{E(\theta)\}_{\theta \in \Theta}$ is a *family of e-variables* for $\{\mathcal{P}_\theta\}_{\theta \in \Theta}$ if, for each θ , $E(\theta)$ is an e-variable for $\mathcal{P}_\theta$, i.e. $\mathbb{E}^\mathbb{P}[E(\theta)] \leq 1$ for every $\mathbb{P} \in \mathcal{P}_\theta$.

$(1 - \alpha)$ -confidence interval (CI)

For fixed $\alpha \in [0, 1]$, a set $C(\alpha) \subseteq \Theta$ is a $(1 - \alpha)$ -CI for ϑ if $\mathbb{P}(\vartheta(\mathbb{P}) \in C(\alpha)) \geq 1 - \alpha$ for all $\mathbb{P} \in \mathcal{P}$. $\{C(\alpha)\}_{\alpha \in [0, 1]}$ is a *family of CIs* if $C(\alpha)$ is a $(1 - \alpha)$ -CI for every α .

Definition 13.1: families of e-variables and e-CIs II

$(1 - \alpha)$ -e-confidence interval (e-CI)

$C(\alpha)$ is a $(1 - \alpha)$ -e-CI if there *exists* a family of e-variables $\{E(\theta, \alpha)\}_{\theta \in \Theta}$ for $\{\mathcal{P}_\theta\}_{\theta \in \Theta}$ such that

$$C(\alpha) = \left\{ \theta \in \Theta : E(\theta, \alpha) < \frac{1}{\alpha} \right\}.$$

$\{C(\alpha)\}_{\alpha \in [0,1]}$ is a *family of e-CIs* if $C(\alpha)$ is an e-CI for every α .

Level-free

A family of e-CIs is *level-free* if the underlying e-variable $E(\theta, \alpha)$ does **not** depend on α – we then write it simply $E(\theta)$, and *one* object $\{E(\theta)\}_\theta$ produces valid CIs at *every* level α at once. This is the interesting, powerful case.

Convention: $C(0) = \Theta$, $C(1) = \emptyset$, and $C(\alpha) \supseteq C(\alpha')$ for $\alpha \leq \alpha'$ (automatic for level-free families).

Two easy facts: $e\text{-CI} \Rightarrow \text{CI}$, and $\text{CI} \Rightarrow e\text{-CI}$ (trivially)

Fact 1: every $e\text{-CI}$ is a CI

For $\mathbb{P} \in \mathcal{P}_\theta$, Markov's inequality gives

$$\mathbb{P}(\theta \notin C(\alpha)) = \mathbb{P}(E(\theta, \alpha) \geq \frac{1}{\alpha}) \leq \alpha \cdot \mathbb{E}[E(\theta, \alpha)] \leq \alpha.$$

(This is also why we used strict “ $< 1/\alpha$ ” rather than “ $\leq$ ”: it yields the smallest valid $e\text{-CI}$.)

Fact 2: every CI is (trivially) an $e\text{-CI}$

Given any $(1 - \alpha)\text{-CI}$ $C(\alpha)$, define the *all-or-nothing* $e\text{-variable}$

$$E(\theta, \alpha) := \frac{1}{\alpha} \mathbb{1}_{\{\theta \notin C(\alpha)\}}.$$

Then $\mathbb{E}[E(\theta, \alpha)] = \frac{1}{\alpha} \mathbb{P}(\theta \notin C(\alpha)) \leq \frac{1}{\alpha} \cdot \alpha = 1$, so it is a valid $e\text{-variable}$, and one checks $\{\theta : E(\theta, \alpha) < 1/\alpha\} = C(\alpha)$ exactly.

This embedding is *not* level-free (E depends on α through both the $1/\alpha$ scaling and the set $C(\alpha)$ itself) – so while every CI “counts” as an $e\text{-CI}$, the genuinely useful $e\text{-CIs}$ are the level-free ones built from a

Example 13.2: an e-CI for a Gaussian mean, with just $n = 1$ observation

A single-observation e-CI

For $X_1 \sim N(\mu, \sigma^2)$ with σ^2 *unknown*, the set

$$X_1 \pm \alpha^{-1} \exp((X_1^2 - 1)/2)$$

is a valid $(1 - \alpha)$ -e-CI for μ , built from the level-free e-variable family $\{|X_1 - \mu| \exp((1 - X_1^2)/2)\}_{\mu \in \mathbb{R}}$, valid for $\bigcup_{\sigma > 0} N(\mu, \sigma^2)$.

Here α = miscoverage level, X_1 = the single observation, μ = the candidate mean being tested. In sharp contrast: the classical Student- t CI for μ needs *at least two* observations to even be well-defined (you need one data point just to estimate the variance). This is a first hint that e-CIs are a genuinely different (not merely repackaged) object – not just “inflated classical CIs.”

13.1.1 E-CIs from stopped confidence sequences I

Intuition. In sequential settings (Chapter 7), data $X_1, X_2, \dots$ arrive one at a time, and we may want a CI that stays valid *no matter when we decide to stop looking* – even if the stopping time is chosen by peeking at the data. A confidence *sequence* formalizes this.

Definition 13.3: confidence sequence

Given $\alpha \in [0, 1]$, a $(1 - \alpha)$ -confidence sequence for ϑ is a sequence of sets $(C^t(\alpha))_{t \in \mathbb{N}}$ such that $C^\tau(\alpha) \subseteq \Theta$ is a $(1 - \alpha)$ -CI *for any stopping time* τ . Equivalently:
 $\mathbb{P}(\vartheta(\mathbb{P}) \in C^t(\alpha) \text{ for all } t \in \mathbb{N}) \geq 1 - \alpha$.

Here τ is a random time (possibly data-dependent, e.g. “stop as soon as I’m bored”) satisfying the usual measurability condition $\{\tau \leq t\} \in \mathcal{F}_t$ with respect to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$ generated by the accumulating data.

13.1.1 E-CIs from stopped confidence sequences II

Construction. For fixed α , let $E(\theta, \alpha) \equiv \{E_t(\theta, \alpha)\}_{t \geq 0}$ be an *e-process* for $\mathcal{P}_\theta$ (the sequential generalization of an e-value from Chapter 7: at every stopping time, evaluating the process there still gives a valid e-value). Then

$$C^t(\alpha) = \left\{ \theta \in \Theta : E_t(\theta, \alpha) < \frac{1}{\alpha} \right\}$$

is a confidence sequence for ϑ , by *Ville's inequality* (the maximal/anytime version of Markov's inequality for nonnegative supermartingales).

Proposition 13.4

If $\{C^t(\alpha)\}_{t \in \mathbb{N}}$ is constructed as above, then for *any* stopping time τ , $C^\tau(\alpha)$ is an e-CI. Furthermore, if $E(\theta, \alpha)$ does not depend on α , then $\{C^\tau(\alpha)\}_{\alpha \in [0,1]}$ is a *level-free* family of e-CIs.

This is the key mechanism: because Ville's inequality controls the whole path of the e-process (not just one fixed time), you may stop – i.e., choose which data-dependent moment to “read off” your interval – *adaptively*, and the guarantee survives.

13.1.2 E-CIs by calibrating CIs I

Intuition. Suppose you already trust some ordinary (non-e) family of nested CIs $\{C(\alpha)\}_\alpha$ (e.g. the exact binomial Clopper–Pearson family, or a Wald interval). Can you mechanically convert it into a genuinely *level-free* e-CI family, without redoing any statistics? Yes – via a *calibrator*.

Calibrator (recap)

A calibrator $f : [0, 1] \rightarrow [0, \infty]$ is a nonincreasing function turning any valid p-value P into a valid e-value $f(P)$ (i.e. $\mathbb{E}[f(P)] \leq 1$ whenever P is a genuine p-value). Its right inverse is $f^{-1}(x) := \sup\{p : f(p) \geq x\}$.

Decreasing / continuous-from-below CI functions

Call $C : [0, 1] \rightarrow 2^\Theta$ *decreasing* if $\alpha \leq \beta \Rightarrow C(\alpha) \supseteq C(\beta)$, and *continuous from below* if $C(\alpha) = \bigcup_{\beta > \alpha} C(\beta)$ for every α . (Nested CI families you meet in practice satisfy both.)

13.1.2 E-CIs by calibrating CIs II

Theorem 13.5: calibrated e-CI

If C is decreasing, continuous from below, and $C(\alpha)$ is always a valid $(1 - \alpha)$ -CI, and f is a calibrator with right inverse f^{-1} , then

$$C^{\text{cal}}(\alpha) := C\left(f^{-1}\left(\frac{1}{\alpha}\right)\right) \supseteq C(\alpha)$$

is a $(1 - \alpha)$ -e-CI, and $\{C^{\text{cal}}(\alpha)\}_{\alpha \in [0,1]}$ is a *level-free* family of e-CIs.

Proof idea. Define the “dual” p-value $P^{\text{dual}}(\theta) := \inf\{\alpha \in [0, 1] : \theta \notin C(\alpha)\}$ – this is a genuine p-value for $\mathcal{P}_\theta$ (it is exactly the usual duality between a nested CI family and a p-value function). Then $E^{\text{cal}}(\theta) := f(P^{\text{dual}}(\theta))$ is an e-variable *not depending on α* (level-free!), and algebra shows $\{\theta : E^{\text{cal}}(\theta) < 1/\alpha\} = C(f^{-1}(1/\alpha))$.

13.1.2 E-CIs by calibrating CIs III

A concrete calibrator. The standard family $f_\kappa(p) := \kappa p^{\kappa-1}$ for $\kappa \in (0, 1)$ satisfies $\int_0^1 f_\kappa(p) dp = 1$, so it is a valid calibrator for every κ . Its inverse is $f_\kappa^{-1}(y) = (\kappa/y)^{1/(1-\kappa)}$, so

$$f_\kappa^{-1}(1/\alpha) = (\kappa\alpha)^{1/(1-\kappa)}.$$

Numerically, for $\alpha = 0.05$ and $\kappa = 1/2$: $f_\kappa^{-1}(1/\alpha) = (0.5 \times 0.05)^2 = 0.000625$. So a calibrated 95% e-CI equals the underlying classical CI at confidence level $1 - 0.000625 = 99.94\%$ – i.e. *much* wider than the classical 95% CI. **This is the price of full level-freeness from a generic calibrator:** getting a family valid at every α simultaneously, from an arbitrary CI family, is intrinsically conservative. (Optimizing κ helps only a little: at $\kappa \approx 0.2$ the same computation gives $\approx 99.68\%$ instead.) Bespoke constructions – like Example 13.2, or the e-process/Ville's-inequality route of Section 13.1.1 – are typically far tighter than this generic recipe.

Python: 13.1 – e-CIs for our running coin example

Six coins are each flipped $n = 20$ times; unknown to us, every coin is secretly fair ($\theta^* = 0.5$). We use exact Clopper–Pearson binomial intervals – which, by the trivial embedding (Fact 2 above), are automatically also e-CIs at each fixed α – as our running numerical example for the rest of the chapter.

```
import numpy as np
from scipy.stats import beta as beta_dist

def clopper_pearson(x, n, conf):
    """Exact (1-alpha)-CI for a Binomial(n,theta) proportion."""
    a = 1 - conf
    lo = 0.0 if x == 0 else beta_dist.ppf(a/2, x, n-x+1)
    hi = 1.0 if x == n else beta_dist.ppf(1-a/2, x+1, n-x)
    return lo, hi

coins = {"A": 15, "B": 5, "C": 11, "D": 9, "E": 8, "F": 12} # heads out of n=20
n, alpha = 20, 0.05
for name, heads in coins.items():
    lo, hi = clopper_pearson(heads, n, 1-alpha)
    phat = heads / n
    print(f"Coin {name}: phat={phat:.2f} 95% e-CI=[{lo:.3f},{hi:.3f}]"
          f" covers 0.5: {lo <= 0.5 <= hi}")

# Coin A: phat=0.75 95% e-CI=[0.509,0.913] covers 0.5: False
# Coin B: phat=0.25 95% e-CI=[0.087,0.491] covers 0.5: False
# Coin C: phat=0.55 95% e-CI=[0.315,0.769] covers 0.5: True
# Coin D: phat=0.45 95% e-CI=[0.231,0.685] covers 0.5: True
# Coin E: phat=0.40 95% e-CI=[0.191,0.639] covers 0.5: True
# Coin F: phat=0.60 95% e-CI=[0.361,0.809] covers 0.5: True
```

13.2.1 Intuition and setup: the goal of FCR control

A scientist observes $\mathbf{X} = (X_1, \dots, X_K)$ and is potentially interested in K functionals $\vartheta_1, \dots, \vartheta_K$, with true values $\theta_i^* := \vartheta_i(\mathbb{P})$. Perhaps they want the $L \geq 1$ largest values of $\boldsymbol{\theta}^*$, or any index exceeding some threshold – in short, *which* indices matter is often only clear *after* seeing the data.

For each i , the scientist can build a $(1 - \alpha)$ -CI $C_i(\alpha)$ from X_i . They then pick a data-dependent subset $S = S(\mathbf{X}) \subseteq \mathcal{K} = \{1, \dots, K\}$ of “interesting” parameters, and must choose confidence levels $\{\alpha_i\}_{i \in S}$ (possibly data-dependent too) for the CIs they actually report.

False coverage proportion (FCP) and false coverage rate (FCR)

$$\text{FCP} := \text{FCP}(S, \{\alpha_i\}_{i \in S}) := \frac{\sum_{i \in S} \mathbb{1}\{\theta_i^* \notin C_i(\alpha_i)\}}{|S| \vee 1}, \quad \text{FCR} := \mathbb{E}^{\mathbb{P}}[\text{FCP}].$$

S : selected index set; α_i : confidence level used for parameter i ; $|S| \vee 1 = \max(|S|, 1)$ avoids division by zero. If every α_i equals a common γ , we write $\text{FCP}(S, \gamma)$.

Goal: choose $\{\alpha_i\}_{i \in S}$ so that $\text{FCR} \leq \delta$ for a target δ , *for every distribution $\mathbb{P}$, and no matter what the selection rule S is* – even if S is unknown to us.

The (classical) Benjamini–Yekutieli procedure

The point of comparison. The BY procedure's validity depends on assumptions about the dependence in $\mathbf{X}$ and about $\mathcal{S}$:

- If $X_1, \dots, X_K$ are **mutually independent** (and $\mathcal{S}$ satisfies some regularity conditions), BY sets

$$\alpha_i = \delta |S|/K, \quad i \in S.$$

- Under **arbitrary dependence** and an unrestricted/unknown selection rule, BY must instead use the more conservative

$$\alpha_i = \frac{\delta |S|}{K \ell_K}, \quad \ell_K := \sum_{i=1}^K \frac{1}{i} \approx \log K$$

(ℓ_K is the K -th harmonic number – e.g. $\ell_6 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} = 2.45$).

This is reminiscent of BH vs. the log-penalized BH procedure under arbitrary dependence (Chapter 9). The natural hope, by analogy with e-BH (which controls FDR under arbitrary dependence *without* the $\log K$ penalty), is that an **e-CI-based** procedure can control FCR under arbitrary dependence *without paying the ℓ_K tax*. That is exactly the e-BY procedure.

Definition 13.6 and Theorem 13.7: the e-BY procedure

Definition 13.6: e-BY procedure

At level $\delta \in [0, 1]$, the e-BY procedure sets

$$\alpha_i = \delta |S|/K \quad \text{for each } i \in S.$$

Same formula as classical BY under independence – the difference is entirely in *which guarantee it satisfies*, and *why*.

Theorem 13.7

Let $\{C_i(\alpha)\}_{\alpha \in [0,1]}$ be a *level-free* family of e-CIs, for each $i \in \mathcal{K}$. Then the e-BY procedure ensures $\text{FCR} \leq \delta$ for any $\delta \in (0, 1)$, under **any** dependence structure between $X_1, \dots, X_K$, and for **any** selection rule $\mathcal{S}$ (even unknown to us). In fact, the stronger simultaneous guarantee holds:

$$\mathbb{E}^{\mathbb{P}} \left[\sup_{S \in 2^{\mathcal{K}}} \sup_{\delta \in (0,1)} \frac{\text{FCP}(S, \delta |S|/K)}{\delta} \right] \leq 1 \quad \text{for all } \mathbb{P} \in \mathcal{P}.$$

Why level-free matters: the proof needs one fixed e-variable $E_i(\theta_i^*)$ per index – not depending on α – so that plugging in the data-dependent threshold $\delta |S|/K$ *after* the fact still preserves $\mathbb{E}[E_i(\theta_i^*)] \leq 1$.

Proof sketch of Theorem 13.7

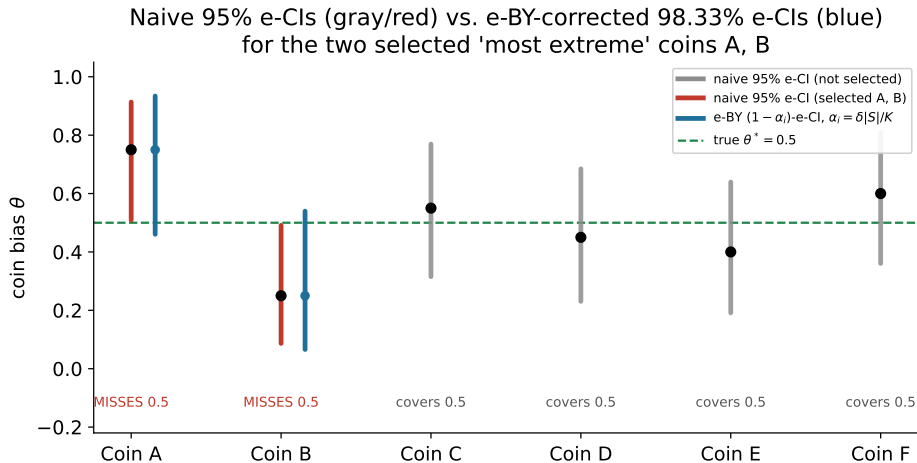
Write $E_i(\theta_i^*)$ for the level-free e-variable behind $C_i(\cdot)$, so $\theta_i^* \notin C_i(\alpha) \iff E_i(\theta_i^*) \geq 1/\alpha$. Then

$$\begin{aligned}\text{FCR} &= \mathbb{E}^{\mathbb{P}} \left[\frac{\sum_{i \in S} \mathbb{1}\{\theta_i^* \notin C_i(\delta|S|/K)\}}{|S| \vee 1} \right] = \mathbb{E}^{\mathbb{P}} \left[\frac{\sum_{i \in \mathcal{K}} \mathbb{1}\{E_i(\theta_i^*)|S|\delta/K > 1\} \cdot \mathbb{1}\{i \in S\}}{|S| \vee 1} \right] \\ &\leq \sum_{i \in \mathcal{K}} \mathbb{E}^{\mathbb{P}} \left[\frac{E_i(\theta_i^*)|S|\delta}{K(|S| \vee 1)} \right] = \sum_{i \in \mathcal{K}} \frac{\delta}{K} \mathbb{E}^{\mathbb{P}} \left[E_i(\theta_i^*) \cdot \frac{|S|}{|S| \vee 1} \right] \leq \sum_{i \in \mathcal{K}} \frac{\delta}{K} = \delta,\end{aligned}$$

using $\mathbb{1}\{x > 1\} \leq x$ for the first inequality, and $\mathbb{E}[E_i(\theta_i^*)] \leq 1$ together with $|S|/(|S| \vee 1) \leq 1$ for the second. No independence or knowledge of S was used anywhere – exactly like the “averaging is always a valid e-merge, regardless of dependence” fact from Chapter 8. **Note:** FCR control by e-BY also implies FDR control by e-BH, but not conversely.

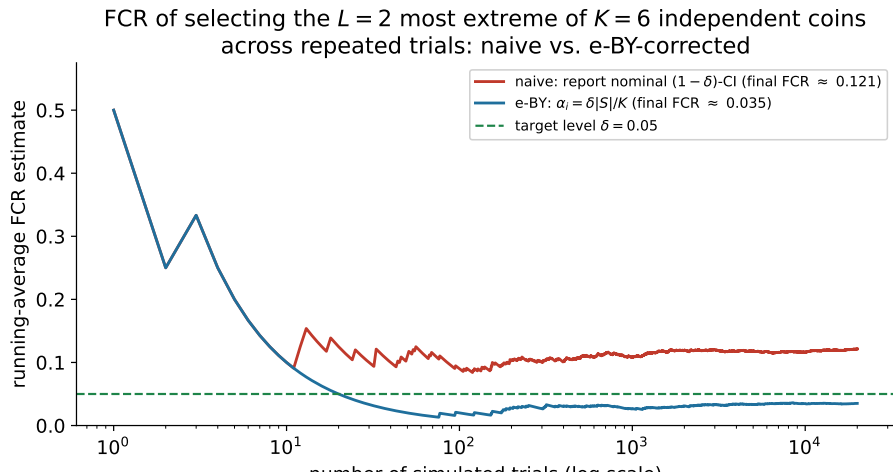
Running example, step 1: naive reporting breaks down

Recall: $K = 6$ coins, $n = 20$ flips each, all secretly fair ($\theta_i^* = 0.5$); observed heads 15, 5, 11, 9, 8, 12 (coins A–F). The scientist's selection rule: report the $L = 2$ *most extreme* observed biases – here, coins A ($\hat{p} = 0.75$) and B ($\hat{p} = 0.25$).



Running example, step 2: how bad is the naive FCR, really?

A single unlucky draw isn't proof of anything – so we simulate: repeat the “6 fair coins, flip 20 times, select the 2 most extreme, report nominal 95% CIs” experiment many times, and track what fraction of the *reported, selected* intervals miss 0.5.



Running example, step 3: applying the e-BY correction

The fix (Definition 13.6): since $|S| = 2$ and $K = 6$, use $\alpha_i = \delta|S|/K = 0.05 \times 2/6 = 1/60 \approx 0.0167$ for the two selected coins – i.e., report $(1 - \frac{1}{60}) \approx 98.33\%$ intervals instead of 95%.

Real adjusted numbers

Coin	naive 95% e-CI	e-BY (98.33%) e-CI	covers 0.5?
A (15/20)	[0.509, 0.913] $\times$	[0.460, 0.934]	yes
B (5/20)	[0.087, 0.491] $\times$	[0.066, 0.540]	yes

Widening from 95% to 98.33% (a modest correction, since $|S|/K = 1/3$ here) is enough to restore coverage in this instance. Over the repeated-trials simulation (previous slide), the **e-BY-corrected empirical FCR** $\approx 0.035 \leq \delta = 0.05$: controlled, as Theorem 13.7 guarantees. And because our coins are independent, this numeric illustration works even with plain (non-e) Clopper–Pearson CIs; the genuine e-BY guarantee (Theorem 13.7) is what lets the *same* formula $\alpha_i = \delta|S|/K$ keep working even if the coins were arbitrarily dependent and S were unknown – unlike classical BY, which would then need the extra $\ell_6 = 2.45$ penalty, i.e. $\alpha_i = 0.05 \times 2/(6 \times 2.45) \approx 0.0068$ (a 99.32% interval – noticeably more conservative).

Python: 13.2 – simulating the FCR of naive vs. e-BY reporting

```
import numpy as np
from scipy.stats import beta as beta_dist

def clopper_pearson(x, n, conf):
    a = 1 - conf
    lo = 0.0 if x == 0 else beta_dist.ppf(a/2, x, n-x+1)
    hi = 1.0 if x == n else beta_dist.ppf(1-a/2, x+1, n-x)
    return lo, hi

rng = np.random.default_rng(2024)
K, n, L, delta, theta_true = 6, 20, 2, 0.05, 0.5
n_trials = 20000
miss_naive = miss_corrected = 0

for _ in range(n_trials):
    x = rng.binomial(n, theta_true, size=K)
    phat = x / n
    sel = np.argsort(-np.abs(phat - 0.5))[:L]
    alpha_i = delta * L / K
    for i in sel:
        lo_n, hi_n = clopper_pearson(x[i], n, 1 - delta)
        lo_c, hi_c = clopper_pearson(x[i], n, 1 - alpha_i)
        miss_naive += not (lo_n <= theta_true <= hi_n)
        miss_corrected += not (lo_c <= theta_true <= hi_c)

print("naive FCR estimate:      ", miss_naive / (n_trials * L))
print("e-BY corrected FCR:      ", miss_corrected / (n_trials * L))
```

13.3 Intuition: many agents, one target, how to merge?

Suppose K different “agents” each hand you an uncertainty set C_k for the *same* target quantity c – e.g. K different point-estimation methods each with their own CI for a parameter, or K different algorithms each producing a conformal prediction set for the same future outcome. The sets may be built very differently and be arbitrarily dependent (shared data, correlated methods).

Goal: combine $C_1, \dots, C_K$, in a black-box manner (only seeing the sets themselves, not how they were built), into one merged set that keeps a valid coverage guarantee while staying reasonably small.

Setup (Eq. 13.1)

Each $C_k = C_k(z)$, $k \in \mathcal{K}$, based on observed data z , satisfies its own marginal guarantee at level $\alpha \in (0, 1)$:

$$\mathbb{P}(c \in C_k) \geq 1 - \alpha, \quad k \in \mathcal{K}.$$

Here c is the (possibly random, e.g. in conformal prediction) target the agents are all trying to cover.

Two trivial (bad) solutions: union and intersection

Union C^J

$$C^J := \bigcup_{k=1}^K C_k.$$

Always respects the coverage guarantee (it only grows the set), but is typically *far too large* – coverage becomes badly inflated (much bigger than $1 - \alpha$) at the cost of a useless, oversized set.

Intersection C^I

$$C^I := \bigcap_{k=1}^K C_k.$$

Narrower, but by the union bound only guarantees $\mathbb{P}(c \in C^I) \geq 1 - K\alpha$ – **uninformative** once K is even moderately large (e.g. $K\alpha \geq 1$).

We want something in between: nearly as small as the individual sets, with a coverage guarantee that doesn't degrade badly with K . The answer, again, runs through e-values.

13.3.1 The majority vote procedure

Majority vote set (Eq. 13.2)

$$C^M := \left\{ s \in \mathcal{S} : \frac{1}{K} \sum_{k=1}^K \mathbb{1}\{s \in C_k\} > \frac{1}{2} \right\}.$$

C^M contains exactly the points s that *more than half* of the K agents' sets agree contain c . ($\mathcal{S}$ is the space in which the target c lives.)

Theorem 13.8

For $K \geq 2$ sets satisfying the coverage property above,

$$\mathbb{P}(c \in C^M) \geq 1 - 2\alpha.$$

Proof. Let $\phi_k := \mathbb{1}\{c \notin C_k\}$, a Bernoulli variable with $\mathbb{E}[\phi_k] \leq \alpha$; then $E_k := \phi_k/\alpha$ is an e-value, and so is $\bar{E} := (E_1 + \dots + E_K)/K$ – by the exact same “averaging is always valid” fact used throughout this book (Chapter 8 for merging e-values; the e-BY proof above). Then

$$\mathbb{P}(c \notin C^M) = \mathbb{P}\left(\frac{1}{K} \sum_k \phi_k \geq \frac{1}{2}\right) = \mathbb{P}(\bar{E} \geq \frac{1}{2\alpha}) \leq 2\alpha$$

How tight is $1 - 2\alpha$, and a general threshold τ

The bound $1 - 2\alpha$ is *worst-case tight*: with K odd and an adversarial joint distribution over the C_k 's (either all agents cover c , or exactly $\lfloor K/2 \rfloor$ do), the miscoverage of majority vote can equal $\alpha K / \lfloor K/2 \rfloor \rightarrow 2\alpha$. But it is often much better in practice: if the C_k are *independent*, Hoeffding's inequality gives miscoverage *exponentially small* in K (not just $\leq 2\alpha$); if the C_k are all identical, coverage is exactly $1 - \alpha$.

General threshold τ (Eq. 13.4) and Theorem 13.10

For any $\tau \in [0, 1)$, define

$$C^\tau := \left\{ s \in \mathcal{S} : \frac{1}{K} \sum_{k=1}^K \mathbb{1}\{s \in C_k\} > \tau \right\}, \quad \mathbb{P}(c \in C^\tau) \geq 1 - \frac{\alpha}{1 - \tau}.$$

Larger $\tau \Rightarrow$ smaller set but weaker guarantee. $\tau = 1/2$ recovers majority vote; $\tau = 1 - 1/K$ recovers the intersection C^I ; $\tau = 0$ recovers the union C^J – one theorem, one dial, interpolating between all our earlier constructions.

Theorem 13.12: majority vote is never more than twice as large

Size bound

Let $\nu(\cdot)$ be a measure of size (Lebesgue measure for intervals, counting measure for discrete sets). Then for all $\tau \in [0, 1)$,

$$\nu(C^\tau) \leq \frac{1}{K^\tau} \sum_{k=1}^K \nu(C_k),$$

and if the input sets are one-dimensional intervals, then for $\tau \in [\frac{1}{2}, 1)$, $\nu(C^\tau) \leq \max_k \nu(C_k)$.

So at $\tau = 1/2$, the merged set's size never exceeds *twice* the average size of the inputs – and if the inputs are intervals, it never exceeds the size of the single *largest* input interval.

Numeric caution – majority vote isn't always the smartest merge. Take three *nested* intervals $C_1 \supset C_2 \supset C_3$ of widths 10, 8, 3. The majority-vote set is C_2 , width 8. But simply picking *one* of the three uniformly at random (which still has exact coverage $1 - \alpha$!) has expected width $(10 + 8 + 3)/3 \approx 7$ – *smaller* than majority vote. Majority vote's real payoff shows up once we also exploit structure like exchangeability, next.

13.3.2 Improving majority vote: exchangeable sets

Intuition. If $C_1, \dots, C_K$ are *exchangeable* (not necessarily independent, but their joint law is permutation-invariant), we can do strictly better than plain majority vote, for free.

Sequential majority vote C^E and Theorem 13.13

Let $C^M(1:k)$ denote the majority vote of just the first k sets. Define

$$C^E := \bigcap_{k=1}^K C^M(1:k) = \left\{ s \in \mathcal{S} : \frac{1}{k} \sum_{j=1}^k \mathbb{1}\{s \in C_j\} > \frac{1}{2} \text{ holds for all } k \in \mathcal{K} \right\}.$$

If $C_1, \dots, C_K$ are exchangeable with coverage $1 - \alpha$ each, then C^E has coverage $\geq 1 - 2\alpha$ and is never worse than plain majority vote: $C^E \subseteq C^M$.

Corollary 13.14: derandomize by random order

If $C_1, \dots, C_K$ are *arbitrarily* dependent, let π be an independent uniformly random permutation and apply C^E to the sets in order $\pi(1), \dots, \pi(K)$: this induces exchangeability, so the resulting C^π is still a $1 - 2\alpha$ set, and still $C^\pi \subseteq C^M$.

Processing sets in a random order and intersecting the running majority votes can only shrink the

Randomized and weighted majority vote

Randomized threshold: C^R (Eq. 13.6) and C^U (Eq. 13.7)

For $U \sim \text{Unif}[0, 1]$ independent of the data, with realization u :

$$C^R := \left\{ s : \frac{1}{K} \sum_k \mathbb{1}\{s \in C_k\} > \frac{1}{2} + \frac{u}{2} \right\}, \quad C^U := \left\{ s : \frac{1}{K} \sum_k \mathbb{1}\{s \in C_k\} > u \right\}.$$

Theorem 13.15

C^R has coverage $\geq 1 - 2\alpha$ and is never larger than C^M ; C^U has coverage $\geq 1 - \alpha$ (matching the nominal input level!) but is never smaller than C^R . The price for C^U 's nominal-level coverage is that it needs external randomization.

Weighted vote C^W (Eq. 13.8) and Theorem 13.16

With prior weights $w_k \in [0, 1]$, $\sum_k w_k = 1$ (fixed *before* seeing the sets, e.g. reflecting trust in agent k):

$$C^W := \left\{ s : \sum_k w_k \mathbb{1}\{s \in C_k\} > \frac{1}{2} + \frac{u}{2} \right\}, \quad \mathbb{P}(c \in C^W) \geq 1 - 2\alpha, \quad \nu(C^W) \leq 2 \sum_k w_k \nu(C_k).$$

Equal weights $w_k = 1/K$ recovers C^R . Proposition 13.17 further allows each agent to have its *own*

Majority vote for confidence sequences and derandomizing estimators

Proposition 13.18 and Theorem 13.19

K different $(1 - \alpha)$ confidence *sequences* tracked in parallel: their majority vote is a $(1 - 2\alpha)$ confidence sequence. If the sequences arrive one-at-a-time (exchangeably, or in random order), the *running* majority vote $C^E(1:t) := \bigcap_{i=1}^t C^M(1:i)$ satisfies

$$\mathbb{P}(\exists t \geq 1 : c \notin C^E(1:t)) \leq 2\alpha.$$

This lets you *derandomize on the fly*: keep merging estimators/CIs as they arrive, watch the sequence stabilize, and stop whenever you like – the guarantee holds for whatever (random, data-dependent) stopping point you chose.

Corollary 13.21: derandomizing point estimators

If $\hat{\theta}_1, \dots, \hat{\theta}_K$ are estimators with $\mathbb{P}(|\hat{\theta}_k - \theta| \leq w(n, \alpha)) \geq 1 - \alpha$, their *median* $\hat{\theta}_{(\lceil K/2 \rceil)}$ satisfies $\mathbb{P}(|\hat{\theta}_{(\lceil K/2 \rceil)} - \theta| \leq w(n, \alpha)) \geq 1 - 2\alpha$ – and this holds simultaneously over all K if the estimators are exchangeable.

13.3.3 Median of midpoints, and derandomizing median-of-means

Theorem 13.20: median of midpoints

If $C_1, \dots, C_K$ are intervals of the *same width* with midpoints $c_1, \dots, c_K$, then the majority vote C^M is contained in (and hence no wider than) the interval centered at the *median* midpoint $c_{(\lceil K/2 \rceil)}$ – a much cheaper computation than the general majority-vote construction, exactly recovering C^M when $\bigcap_k C_k \neq \emptyset$.

Corollaries 13.21–13.22: median-of-median-of-means (MoMoM)

The median-of-means (MoM) estimator $\hat{\mu}^{\text{MoM}}$ (splitting n points into B buckets, averaging within each, then taking the median across buckets) gives sub-Gaussian tails but is *randomized* by the random bucketing. Repeating MoM independently K times and taking the *median of the K resulting estimates* (“MoMoM”) derandomizes it, at the mild cost of doubling the deviation probability:
 $\mathbb{P}(\exists K \geq 1 : |\hat{\mu}_K^{\text{MoMoM}} - \mu| \geq C\sigma\sqrt{t/n}) \leq 4e^{-t}$ (vs. $2e^{-t}$ for one run of MoM) – while the simultaneity over K means you can watch the estimate stabilize and stop adaptively.

13.3.4 Empirical examples from the book

Lasso conformal prediction. $K = 20$ conformal prediction intervals (one per tuning parameter λ) merged with equal weights $w_k = 1/K$ at $\alpha = 0.05$: empirical coverage of C^M was 0.97, of the smaller C^R was 0.92 – both comfortably above the guaranteed $1 - 2\alpha = 0.90$, and the randomized-permutation version C^π achieved 0.93 coverage with smaller sets than plain majority vote.

Parkinson's UPDRS prediction (real data, $K = 4$ algorithms)

	LM	Lasso	RF	NN	C^M	C^R	C^U	C^π
Coverage	0.958	0.960	0.949	0.961	0.951	0.923	0.961	0.918
Width	40.1	40.2	13.3	32.5	29.5	20.6	32.5	20.7

All merges achieve good coverage; C^R and C^π are markedly narrower than the raw linear-model / lasso / neural-net intervals, at only a modest coverage cost relative to the (already excellent) random forest. Since random forest performs best individually here, upweighting it in C^W would shrink the merged set further.

Python: 13.3 – majority vote on a small numeric example

```
import numpy as np

# 5 arbitrarily-dependent intervals for the same target c, each with
# marginal coverage >= 1-alpha (alpha = 0.10 here, so 1-2*alpha = 0.80 guaranteed)
C = [(1.0, 6.5), (2.0, 6.0), (2.3, 5.0), (2.6, 7.0), (3.4, 4.6)]    # (lo, hi)
K = len(C)

def majority_vote(intervals, tau=0.5, n_grid=200000, lo_grid=0.0, hi_grid=8.0):
    """Approximate  $C^\tau$  by evaluating the vote fraction on a fine grid."""
    s = np.linspace(lo_grid, hi_grid, n_grid)
    votes = np.zeros_like(s)
    for lo, hi in intervals:
        votes += (s >= lo) & (s <= hi)
    in_set = (votes / K) > tau
    if not in_set.any():
        return None
    pts = s[in_set]
    return float(pts.min()), float(pts.max())

print("Majority vote  $C^M$  (tau=1/2):", majority_vote(C, tau=0.5))
# -> approx (2.3, 6.0): the region at least 3 of the 5 intervals agree on

# Median-of-midpoints shortcut (Theorem 13.20), valid since here all
# intervals happen to have the same width after recentering by hand:
widths = [hi - lo for lo, hi in C]
print("interval widths:", widths)    # not all equal in general;
                                     # the shortcut needs equal widths
```

Chapter 13 summary I

- An **e-CI** is a confidence set built by inverting a family of e-variables, $C(\alpha) = \{\theta : E(\theta, \alpha) < 1/\alpha\}$; when the underlying e-variable doesn't depend on α (*level-free*), one object gives valid CIs at *every* level simultaneously. Every CI is trivially an e-CI, and every e-CI is a genuine CI (Markov's inequality).
- Level-free e-CIs can be built from stopped confidence sequences / e-processes (Ville's inequality; Section 13.1.1), or by *calibrating* an existing nested CI family with a calibrator (Theorem 13.5; Section 13.1.2) – though generic calibration is noticeably conservative.
- **FCR** is the CI-analogue of FDR (Chapter 9): reporting CIs only for a data-dependently *selected* subset of parameters can badly break the nominal coverage, exactly as reporting only “significant” tests breaks FWER/type-I control.
- The **e-BY procedure** (Definition 13.6) sets $\alpha_i = \delta|S|/K$ for the selected indices; using level-free e-CIs, Theorem 13.7 guarantees $\text{FCR} \leq \delta$ under *arbitrary* dependence and an *arbitrary/unknown* selection rule – unlike classical BY, which needs independence (or an extra $\log K$ -type penalty ℓ_K) for the analogous guarantee.

Chapter 13 summary II

- Our running numeric example: 6 coins, naive 95% CIs for the 2 “most extreme” miss the truth in simulation about 12% of the time; the e-BY correction ($\alpha_i = 1/60$, i.e. 98.33% CIs) brings this down to about 3.5%, safely below the $\delta = 0.05$ target.
- **Majority vote** merges K arbitrarily-dependent uncertainty sets for a shared target into one set with coverage $\geq 1 - 2\alpha$, using at most twice the average size of the inputs – with exchangeable, randomized, and weighted variants that only ever shrink the merged set while preserving the guarantee.